THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 9: **Symmetry** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 5	Number of sides $= 360 \div a$.
2	**(d)** 45	Divide.
3	**(a)** 2	Letters H, I, O, X.
4	**(a)** 150	Add the smallest angles.
5	**(b)** 3	If a figure has a line and order n, it has n lines.
6	**(a)** 3	Order n gives $n - 1$ such turns before a full turn.
7	**(c)** 9	$360 \div$ angle.
8	**(c)** 42	Number of sides times side.
9	**(a)** Both A and R are true and R is the correct explanation of A.	A square is a regular polygon with 4 sides.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true but the order of rotation does not explain the number of lines.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Sides $= 360 \div 30 = 12$ (1); Perimeter $= 12 \times 4 = 48$ cm (1).	1+1
12	Order $= 360 \div 40 = 9$ (1); 120° is a multiple: 3 times 40 (1).	1+1
13	Smallest angle $= 360 \div 4 = 90^\circ$ (1); Angles: 90°, 180°, 270° (1).	1+1
14	Sides $= 360 \div 36 = 10$ (1); Perimeter $= 10 \times 5 = 50$ cm (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Angle $= 360 \div 5 = 72^\circ$ (1); Lines of symmetry $= 5$ (1); Total around centre $= 5 \times 72 = 360^\circ$ (1).	1+1+1
16	Lines through opposite vertices: $4 \div 2 = 2$ (1); Lines through midpoints of opposite sides: 2 (1); Total $= 4$ (1).	1+1+1
17	Smallest angle $= 360 \div 6 = 60^\circ$ (1); $180 \div 60 = 3$ rotations (1); A whole number means the figure fits (1).	1+1+1
	OR: Angle $= 360 \div 6 = 60^\circ$ (1); Lines of symmetry $= 6$ (1); Total around centre $= 6 \times 60 = 360^\circ$ (1).	1+1+1
18	Lines through opposite vertices: $6 \div 2 = 3$ (1); Lines through midpoints of opposite sides: 3 (1); Total $= 6$ (1).	1+1+1
	OR: Smallest angle $= 360 \div 8 = 45^\circ$ (1); $135 \div 45 = 3$ rotations (1); A whole number means the figure fits (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) 6 (1). (ii) 6 (1). (iii) $360^\circ \div 6 = 60^\circ$ (1). (iv) Yes after 180°, because $180^\circ = 3 \times 60^\circ$ (1); no after 100°, because 100 is not a multiple of 60 (1).	1+1+1+2
20	(a) 8 (1); (b) 45° (1); (c) 32 cm (1); (d) 225° (2).	1+1+1+2
	OR: Angles: 90° and 60° (1); Difference 30° (1); LCM of 90 and 60 is 180 (1); Common turn 180° (2).	1+1+1+2

Mistakes to avoid:

- Saying a rectangle has 4 lines of symmetry (diagonals are not lines of symmetry); a non-square rectangle has only 2.
- Giving a parallelogram 2 lines of symmetry; it has none, though it has rotational symmetry of order 2.
- Mixing up the order of rotation with the smallest angle: angle $= 360^\circ \div$ order.
- Saying a figure with order 1 has rotational symmetry; order 1 means only the trivial full turn.
- Drawing the mirror image at the wrong distance from the mirror line or on the same side of it.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 9: **Symmetry** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 40	Divide.
2	**(c)** 2	Letters H, I, O, X.
3	**(c)** 135	Add the smallest angles.
4	**(b)** 5	If a figure has a line and order n, it has n lines.
5	**(c)** 5	Order n gives $n-1$ such turns before a full turn.
6	**(d)** 12	$360 \div$ angle.
7	**(b)** 40	Number of sides times side.
8	**(c)** 16	Equal distance on the other side.
9	**(c)** A is true but R is false.	It fits onto itself after a half turn, so A is true. It has no line of symmetry, so R is false.
10	**(d)** A is false but R is true.	H has two lines of symmetry (vertical and horizontal), so A is false. R is true.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Order $= 360 \div 45 = 8$ (1); 135° is a multiple: 3 times 45 (1).	1+1
12	Smallest angle $= 360 \div 5 = 72^\circ$ (1); Angles: 72°, 144°, 216°, 288° (1).	1+1
13	Sides $= 360 \div 24 = 15$ (1); Perimeter $= 15 \times 6 = 90$ cm (1).	1+1
14	Order $= 360 \div 60 = 6$ (1); 240° is a multiple: 4 times 60 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Angle $= 360 \div 8 = 45^\circ$ (1); Lines of symmetry $= 8$ (1); Total around centre $= 8 \times 45 = 360^\circ$ (1).	1+1+1
16	Lines through opposite vertices: $8 \div 2 = 4$ (1); Lines through midpoints of opposite sides: 4 (1); Total $= 8$ (1).	1+1+1
17	Smallest angle $= 360 \div 9 = 40^\circ$ (1); $200 \div 40 = 5$ rotations (1); A whole number means the figure fits (1).	1+1+1
	OR: Angle $= 360 \div 9 = 40^\circ$ (1); Lines of symmetry $= 9$ (1); Total around centre $= 9 \times 40 = 360^\circ$ (1).	1+1+1
18	Lines through opposite vertices: $10 \div 2 = 5$ (1); Lines through midpoints of opposite sides: 5 (1); Total $= 10$ (1).	1+1+1
	OR: Smallest angle $= 360 \div 10 = 36^\circ$ (1); $144 \div 36 = 4$ rotations (1); A whole number means the figure fits (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) A, H and T (1). (ii) H and S (1). (iii) S (1). (iv) H (1); it has 2 lines of symmetry (vertical and horizontal) and rotational symmetry of order 2 (1).	1+1+1+2
20	(a) 10 (1); (b) 36° (1); (c) 60 cm (1); (d) 108° (2).	1+1+1+2
	OR: Angles: 60° and 40° (1); Difference 20° (1); LCM of 60 and 40 is 120 (1); Common turn 120° (2).	1+1+1+2

Mistakes to avoid:

- Saying a rectangle has 4 lines of symmetry (diagonals are not lines of symmetry); a non-square rectangle has only 2.
- Giving a parallelogram 2 lines of symmetry; it has none, though it has rotational symmetry of order 2.
- Mixing up the order of rotation with the smallest angle: angle $= 360^\circ \div$ order.
- Saying a figure with order 1 has rotational symmetry; order 1 means only the trivial full turn.
- Drawing the mirror image at the wrong distance from the mirror line or on the same side of it.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 9: **Symmetry** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 4	Letters H, I, O, X.
2	**(a)** 162	Add the smallest angles.
3	**(d)** 6	If a figure has a line and order n, it has n lines.
4	**(a)** 7	Order n gives $n-1$ such turns before a full turn.
5	**(d)** 15	$360 \div$ angle.
6	**(c)** 36	Number of sides times side.
7	**(c)** 24	Equal distance on the other side.
8	**(b)** 9	Number of sides $= 360 \div a$.
9	**(a)** Both A and R are true and R is the correct explanation of A.	$360 \div 6 = 60$.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true, but neither explains the other.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Smallest angle $= 360 \div 6 = 60^\circ$ (1); Angles: 60°, 120°, 180°, 240°, 300° (1).	1+1
12	Sides $= 360 \div 20 = 18$ (1); Perimeter $= 18 \times 3 = 54$ cm (1).	1+1
13	Order $= 360 \div 72 = 5$ (1); 216° is a multiple: 3 times 72 (1).	1+1
14	Smallest angle $= 360 \div 8 = 45^\circ$ (1); Angles: 45°, 90°, 135°, 180°, 225°, 270°, 315° (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Angle $= 360 \div 10 = 36^\circ$ (1); Lines of symmetry $= 10$ (1); Total around centre $= 10 \times 36 = 360^\circ$ (1).	1+1+1
16	Lines through opposite vertices: $12 \div 2 = 6$ (1); Lines through midpoints of opposite sides: 6 (1); Total $= 12$ (1).	1+1+1
17	Smallest angle $= 360 \div 12 = 30^\circ$ (1); $210 \div 30 = 7$ rotations (1); A whole number means the figure fits (1).	1+1+1
	OR: Angle $= 360 \div 12 = 30^\circ$ (1); Lines of symmetry $= 12$ (1); Total around centre $= 12 \times 30 = 360^\circ$ (1).	1+1+1
18	Lines through opposite vertices: $14 \div 2 = 7$ (1); Lines through midpoints of opposite sides: 7 (1); Total $= 14$ (1).	1+1+1
	OR: Smallest angle $= 360 \div 5 = 72^\circ$ (1); $288 \div 72 = 4$ rotations (1); A whole number means the figure fits (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) 1 (1). (ii) 0 (1). (iii) 2 (1). (iv) The equilateral triangle, with 3 lines (1); the rectangle has only 2, so the triangle has 1 more (1).	1+1+1+2
20	(a) 5 (1); (b) 72° (1); (c) 40 cm (1); (d) 288° (2).	1+1+1+2
	OR: Angles: 120° and 72° (1); Difference 48° (1); LCM of 120 and 72 is 360 (1); Common turn 360° (2).	1+1+1+2

Mistakes to avoid:

- Saying a rectangle has 4 lines of symmetry (diagonals are not lines of symmetry); a non-square rectangle has only 2.
- Giving a parallelogram 2 lines of symmetry; it has none, though it has rotational symmetry of order 2.
- Mixing up the order of rotation with the smallest angle: angle $= 360^\circ \div$ order.
- Saying a figure with order 1 has rotational symmetry; order 1 means only the trivial full turn.
- Drawing the mirror image at the wrong distance from the mirror line or on the same side of it.

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